

## Rukun (Eric) Qiao

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### Profile

**Hands-on machine learning engineer** with end-to-end experience in robotics and 3D vision. **PhD research** centered on multi-view 3D reconstruction, sensor fusion, and deep learning for geometric perception, specialized in **predictive modeling and transformer model for data-driven decision making**. **Strong software engineering skills** across algorithms, application development, and data workflows; write clean, maintainable code and deliver reliably with testing and automation. **Full-stack developing experience** to move ideas from prototype to production. **Proactive engineer with an analytical mindset** who turns ambiguous problems into clear metrics, dashboards, and experiments, and communicates insights to guide data-driven decisions. **Passionate about using data and machine learning to deliver measurable improvements in business performance and user experience.**

### Technical Skills

- ♦ **Languages:** Python, JavaScript, C++, C#
- ♦ **GenAI:** LLMs, Retrieval-Augmented Generation (RAG)
- ♦ **Machine Learning:** PyTorch, TensorFlow, Scikit-learn
- ♦ **Data Platform:** PostgreSQL, VectorDB, Databricks, Spark
- ♦ **Experimentation:** A/B Test, Power Analysis, Statistic Tests
- ♦ **Full-stack Development:** React Native, FastAPI
- ♦ **DevOp & Cloud:** Git, Docker, Github Actions, Azure, AWS
- ♦ **Data Visualization:** Matplotlib, MATLAB, OpenCV, Open3D

### Soft Skills

- ♦ Agile development
- ♦ Ownership mindset
- ♦ Learning agility
- ♦ Problem solving
- ♦ Coaching & mentoring
- ♦ Analytical mindset
- ♦ Business argument

### Professional Experience

#### Independent Developer | Household Inventory Management App

Mar. 2025 – Now

*Tech Stack: React Native, FastAPI, PostgreSQL, Github Actions, Azure, LLMs, RAG, VectorDB*

- ♦ Engineer a scalable full-stack solution with front end, back end and PostgreSQL persistence.
- ♦ Develop predictive models that learn usage patterns to forecast purchase cycles, restock suggestions, and automatic item notation.
- ♦ Implement GPT-based generative AI and RAG pipelines to power natural-language item entry.
- ♦ Design in-app dashboards and regular reports, paired with backend metrics to oversee resource usage, performance, and cost efficiency.
- ♦ Automate CI/CD with GitHub Actions, deploying to Azure Functions & Static Web Apps.
- ♦ Use JIRA to manage the project roadmap and ensure agile development.

#### Software Engineer | CLCN

Mar. 2025 – Now

*Tech Stack: React Native, Github Actions, Figma*

- ♦ Contribute to the front-end development of an internal ERP web app that creates a transparent, measurable collaboration framework and standardizes handoffs.
- ♦ Translate offline processes into configurable, step-based workflows with deadlines, reminders, document tracking, and clear UI flows/specs aligned with business and design.
- ♦ Build collaboration features around volunteer profiles and availability so organizers can assemble teams, send invitations, and coordinate tasks transparently.
- ♦ Establish unit and component tests for critical flows, define QA checklist and regression suites.
- ♦ Instrument data capture with data tracking to support post-event reviews and training, laying the foundation for analytics-driven improvements and policy updates.

## Machine Learning Engineer | SenseTime

Sep. 2020 – Sep. 2021

*Tech Stack: Python, PyTorch, OpenCV, NumPy, CUDA, Git*

- ♦ Built an automated multi-sensor data pipeline for smartphone 3D capture – standardizing images, time-aligning IMU data, and extracting photometric features – to power downstream 3D reconstruction models at scale.
- ♦ Acted as a professional consultant, engaging stakeholders for requirement collection and success-metric alignment, setting acceptance thresholds and release gates to app performance.
- ♦ Developed a deep-learning multi-view model to reduce reconstruction uncertainty, and enabled a post-deployment fine-tuning loop on user-returned images.
- ♦ Optimized inference efficiency by integrating classical 3D matching algorithms with deep learning models, reducing computational load for mobile deployment.
- ♦ Delivered the solution as a modular Python package with clear docs and tests, and exposed a lightweight REST/HTTP wrapper to streamline integration with the mobile app and internal tooling, enabled downstream 3D-model-based features.

## Computer Vision Engineer | BOE Technology

Sep. 2018 – Sep. 2019

*Tech Stack: C++, OpenCV, MATLAB, ROS*

- ♦ Coordinated cross-functional teams to develop a visual navigation system for autonomous cleaning robots, aligning perception, planning and hardware modules.
- ♦ Installed and calibrated stereo cameras and ancillary sensors, performing intrinsic/extrinsic calibration to achieve sub-millimetre pose accuracy.
- ♦ Implemented C++ modules for real-time sensor data acquisition and preprocessing, improving data reliability under varying lighting conditions, integrated into the commercial cleaning-robot line.
- ♦ Conducted workshops and produced SOPs enabling staff to interpret outputs and perform controlled parameter adjustments for different environments.

## Education

### Peking University | PhD in Artificial Intelligence

*School of Intelligence Science and Technology*

**Beijing, China**

Sep. 2016 – Jul. 2024

### Kyushu University | Visiting Scholar Program

*Graduate School of Information Science and Electrical Engineering*

**Fukuoka, Japan**

Sep. 2018 – Mar. 2019

### Peking University | BSc in Artificial Intelligence

*School of Electronic Engineering and Computer Science*

**Beijing, China**

Sep. 2012 – Jul. 2016

## Research Achievements

### Research Highlights

- ♦ Developed a disparity estimation network leveraging temporal correlation in structured light systems, reducing computing cost by about 50% for robotic vision applications.
- ♦ Pioneered the integration of neural implicit functions for structured light systems, enhancing depth estimation accuracy with minimal input patterns, and provide insightful research direction for structured light researches.

### Publications

- ♦ **Rukun Qiao**, Hiroshi Kawasaki and Hongbin Zha, "TIDE: Temporally Incremental Disparity Estimation via Pattern Flow in Structured Light System," in *IEEE Robotics and Automation Letters*, pp.5111-5118, April 2022.
- ♦ **Rukun Qiao**, Hiroshi Kawasaki and Hongbin Zha, "Online Adaptive Disparity Estimation for Dynamic Scenes in Structured Light Systems," in *IEEE/RSJ International Conference on Intelligent Robotics and Systems*, 2023.
- ♦ **Rukun Qiao**, Hiroshi Kawasaki and Hongbin Zha, "Depth Estimation in Structured Light Systems via Neural Implicit Functions," in *International Conference on 3D Vision*, 2024.